

Enhancing Multimodal Representation Learning for Unpaired Data

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A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science and Engineering

Department of Computer Science and Engineering
Brac University
October, 2025

Declaration

It is hereby declared that

1. The thesis submitted is our own original work while completing degree at Brac University.
2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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Abstract

Multimodal representation learning has become a keystone in the development of artificial intelligence systems that comprehend and integrate data from multiple modalities, including language and vision. Although, current vision-language models are largely dependent on contrastive learning with paired visual-text data, which limits their scalability and their ability to utilize an enormous amount of unpaired data. Moreover, contrastive methods are prone to sample bias brought on by augmentation, which hinders standardization in subsequent tasks. This study proposes a robust and unified multimodal learning framework that seamlessly aligns and fuses representation across modalities. The proposed method aims to improve flexibility, efficiency and generalization by reducing augmentation-related biases while effectively integrating unpaired multimodal data. Through this approach, the study hopes to advance multimodal models that are capable of learning from a range of real-world datasets.

Keywords: Multimodal representation learning, Unpaired data, Vision-language model, Contrastive learning, Cross-modal fusion, Feature fusion, Data-efficient learning.

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Chapter 1

Introduction

This chapter lays the groundwork for investigating enhancements in multimodal representation learning for unpaired data. It begins by exploring the recent methods of LMMs (Large Multimodal Models) and their transformative applications in explainable AI, autonomous driving, and education, while highlighting critical limitations in current methods on expensive, manually curated parallel datasets. The substantial divergence between the reality of billions of mismatched multimodal data points produced every day on the internet and how AI systems are trained using precisely matched image-text pairings serves as the motivation being this research. To address this, the research examines training methods and architectural modifications focused on three key areas: learning from truly unpaired data sources and enhancing robustness across a range of distributions. An experimental methodology is used, which combines rigorous validation with iterative refinement, baseline analysis, and enhancement development and systematic evaluation across diverse dataset types. Lastly, the chapter acknowledges both the need to develop a unified multimodal framework as well as its inherent drawbacks, such as adversarial vulnerability, interpretability issues, and the difficulties of aligning multiple modalities with noisy, unpaired data.

1.1 Background

The widespread use of Large Language Models (LLMs) and their multimodal equivalents, Large Multimodal Models (LMMs), transformed education, content-creation and human-AI interaction. These systems improve learning interfaces and educational methods by providing seamless interaction across several modalities, such as texts, graphs and audio-visual content[1]. However, adversarial manipulation that threatens reliability, privacy, and validity may continue to affect LLMs and LMMs. The significance of scalable multimodal learning approaches that involve several input modalities while retaining resilience and flexibility – especially in unpaired data environments – has been highlighted with this growing concern.

In addition, multimodal deep learning methodologies are becoming increasingly useful in the important field of autonomous driving. Large-foundation models are widely used in end-to-end driving systems to integrate language and vision for per-

ception and control, improving adaptability in large-scale scenarios [2]. Through the utilization of transformer-based spatial feature extraction and pixel-aligned descriptors, these systems exhibit the possibility of multimodal fusion in practical, safety-relevant scenarios. This shows how foundational multimodal models can be used for robust policy learning and decision-making, going beyond image-text alignment and providing insights that can be applied to other unpaired multimodal learning environments.

Furthermore, explainability has become a vital area of research for dealing with deep learning’s ”black-box” nature. Multimodal and multitask deep learning classifiers have been interpreted and improved using eXplainable AI (XAI) techniques in network traffic classification [3]. The relationship between model complexity, input signals and classification performance can be demonstrated by XAI methods, which analyze global interpretation rather than instance-based explanations. This shows the importance of creating not only high-performing multimodal systems but also focusing on how reliable and trustworthy they are in case of decision-making.

In combination, current multimodal learning methods often rely on contrastive learning with separate encoders and paired image-text datasets, which hinders scalability to unpaired real-world data and creates sampling bias. Moreover, existing systems suffer from lack of robustness in open environments and show vulnerability to adversarial attacks. In the midst of these challenges, this thesis proposes a unified and scalable multimodal framework that is capable of jointly aligning and fusing representations from both paired and unpaired data. This approach fills in the gaps found in the domains of explainable AI, autonomous driving and education while improving robustness against adverse manipulations and providing a more flexible foundation for reliable and adaptive multimodal systems.

1.2 Rational of the Study or Motivation

The problems that lie in training current AI systems are the real motivation for studying multimodal unparallel datasets. Nowadays, almost all of the AI models need perfectly aligned data, like images with their respective descriptions. Aligning multimodal data manually is expensive and often impossible at scale.

Currently, researchers use carefully organized datasets like MS-COCO, where each image has five hand-written captions that explain it perfectly. Creation of these datasets costs a huge amount of money and as well as time. For example, a dataset consisting of a million images can cost hundreds of thousands of dollars and take many months to work. Even these perfect datasets represent only a small portion of all the information available on the internet, as most real-world data is not organized.

The significance of studying unparallel datasets is immense for several practical reasons, such as enormous amounts of new images, videos, and texts released on the internet every day, and impossible to manually label or organize all this data. As a result, millions of dollars can be saved if AI can learn from unorganized, unmatched data. The workload of labelling data manually will be gone. Also it can keep improving automatically without constant human help. A self driving car need to

work with imperfect sensor data and unclear road conditions so that it can solve real-world problems. Similarly, AI models trained on labelled, organized data often fail when they experience real-world problems. Moreover, mitigating the need for expensive datasets results in wider access to people. Researchers, small companies can create powerful models using available data on the internet.

Successfully working with unparallel datasets would impact any applications heavily. More smarter, intelligent recommendation systems, better accessibility tools for people with disabilities, effective educational platforms, and AI assistants that truly understand how humans align visual and written information. This research is not just theoretically fascinating but also practically necessary for building AI to acquire human-like intelligence that can work in the real world, where data is naturally unlabelled and unorganized.

1.3 Problem Statement

Multimodal Machine Learning focuses on building models that can interpret and relate information from different modalities, such as language, vision and audio [4]. Traditional multimodal learning approaches are mainly based on parallel datasets, where each data point in a single modality has a direct, manually validated relationship with data points in different modalities. For example, datasets such as MS-COCO and Flickr30k give images with precisely annotated descriptions, creating distinct one-to-one correspondences between visual and textual data. However, to create these parallel datasets it takes a huge manual annotation effort, which costs both time and expense and also restricts their ability to manage the huge amounts of multimodal data generated each day online [5].

This research focuses on the following issues: How can multimodal AI models effectively learn meaningful cross-modal representations from unpaired data sources where direct correspondence between modalities is weak, noisy, or there are no direct relations, while preserving computational efficiency and ensuring robust generalization? However, this research problem is essential due to the fact that a wide amount of practical multimodal data exists in an unparalleled form. However, cutting-edge methods like CLIP need weeks to train on 400 million pairs using massive GPU clusters for weeks restricting accessibility to well-resourced organizations. Additionally, there are three interrelated challenges in this research problem: firstly, the alignment problem, where connections between modalities are fundamentally uncertain. Secondly, the computational efficiency problem, which involves prohibitive resources that exclude smaller institutions. Lastly, the generalization problem, where models suffer accuracy degradation on inconsistent real-world data.

1.4 Objective

This research aims to improve multimodal representation learning for unpaired data by investigating architectural modifications and training strategies that enhance generalization while resolving issues with computational inefficiency in more recent approaches, identifying their limitations, and addressing opportunities for ad-

vancement in multimodal machine learning. This work focuses on three topics that are interrelated and collectively contribute to enhancing multimodal representation learning:

- To ensure that models can efficiently learn cross-modal associations without compromising generalization performance, research and develop lightweight architectural modifications that maintain representational capabilities while improving computational viability.
- To design training strategies that enable models to learn from truly unpaired data, where images and text originate from completely different sources with no inherent relationship, thus enhancing their ability to generalize in real-world, unaligned scenarios.
- To develop strong training frameworks that enhance generalization in a range of data distributions and noise levels, with an emphasis on learning representations that remain transferable and stable even data misalignment and ambiguity.
- To minimize computational hurdles and increase the applicability of sophisticated multimodal methodologies for research and deployment in resource-limited environments, hence promoting accessible multimodal learning.
- To examine a model’s ability to generalize across diverse, unpaired multimodal datasets and propose new assessment techniques that do so more precisely.

1.5 Methodology in Brief

This research uses an experimental methodology that combines comparative analysis, architecture design, and empirical evaluation to explore enhancements in multimodal representation learning for unpaired data. The method sets a strong emphasis on recognizing how various design decisions impact model generalization and computing needs in a range of unpaired data settings. To represent how multimodal learning architectures handle unpaired data, the work begins by placing them into practice.

This research begins by implementing current multimodal learning architectures to understand their mechanisms for handling unpaired data. This includes examining contractive learning formulations [6], dual-encoder frameworks utilized in vision-language models [7], and identifying architectural components that contribute the most to cross-modal understanding. Focusing on lightweight encoding strategies, developing training procedures for truly unpaired data where there is no correspondence signal exists, and putting in place reliable loss functions effective under high noise conditions. Enhanced approaches are evaluated across multiple criteria, including generalization across diverse data distribution, robustness to noise levels and data misalignment, and interpretability of learned correspondences. The findings are examined to understand which enhancements contribute most to improved generalization. This entails comparative analysis across methods, looking at failure cases, and examining trade-offs between efficiency and representation quality.

Standard benchmarks, web-scraped data with weak correspondences, and deliberately unpaired datasets having text and images from separate sources are all used in the experiments.

The research utilizes PyTorch for implementation with code that are publicly available as open-source. Validation incorporates cross-dataset evaluation, ablation research to isolate enhanced contributions, and statistical analysis to ensure solid conclusions about genuine enhancement in multimodal learning from unpaired data.

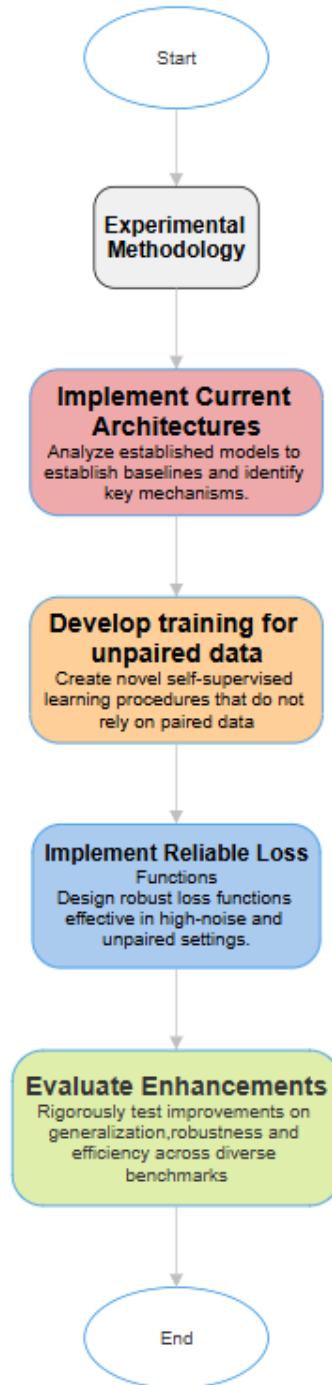


Figure 1.1: Overview of the proposed methodology pipeline for learning from unpaired multimodal data.

1.6 Scopes and Challenges

This research aims to create a unified multimodal learning framework that is capable of using unpaired data to interpret vision and language. The existing methods heavily rely on paired datasets and contrastive objectives. Therefore, the research focuses on achieving joint alignment and fusion in a more flexible and data-efficient way than current approaches. This direction can open up ways to expand multimodal learning beyond selective benchmarks and apply it to a variety of real-world tasks [1].

Despite the scopes, this research encounters a few challenges. Firstly, there are doubts about the accuracy and credibility of multimodal systems in critical domains because they are usually vulnerable to biases and adversary input [2]. Secondly, huge functional models show admirable performance, but are still black-boxes and have difficulty in open-set scenarios, which limits their interpretability and accuracy [3]. Moreover, merging several modalities increases the risks of unstable alignment and uneven representation, especially while working with noisy unpaired data [8].

Addressing these constraints is important for building a scalable, transparent, and reliable multimodal framework that is capable of generalizing efficiently across domains.

Chapter 2

Literature Review

This chapter lays out the theoretical foundations and surveys the state of the art to enhance multimodal representation learning for unpaired data. Essential foundation information on multimodal learning principles is given in the first section of the chapter, examining how models integrate diverse data modalities like text, images, and audio to create a coherent representation. It distinguishes between feature-level and semantic-level alignment. Additionally, it reviews architectural foundations underlying state-of-the-art models like CLIP and ALIGN, critically analyzing their reliance on extensive paired datasets. Furthermore, it explores the crucial trade-off between generalization and computing efficiency in multimodal systems, as well as challenges of learning from unpaired data where explicit correspondences are absent. After that in the section section, a comprehensive analysis of 21 exemplary research papers from a variety of fields was performed, such as healthcare data fusion, knowledge graph completion, cross-modal retrieval, autonomous driving, and medical imaging. Recurring patterns emerge in this thorough review: pervasive reliance on paired or weakly-paired datasets, the computing demands of large-scale models, challenges in handling modality and label inconsistencies, and a restricted capacity to extend to fully unpaired scenarios.

2.1 Preliminaries

This chapter provides the theoretical foundation and essential background necessary to understand the proposed study on enhancing multimodal representation learning for unpaired data. It reviews the key concepts, models, and learning paradigms that form the basis of multimodal learning, focusing on representation alignment, generalization, and computational efficiency.

2.1.1 Fundamentals of Multimodal Learning

The process of integrating information from multiple data modalities, such as text, images, audio, and video, to form cohesive and meaningful representations is known as multimodal learning. The complementary perspectives offered by each modality allow rich semantic understanding and enhance performance on a range of downstream tasks. Aligning diverse data sources that differ in scale, structure, and se-

mantics is the main challenge. Effective multimodal representation learning seeks to map different modalities into a common latent space so that inputs that are semantically related are closely aligned regardless of the modality.

2.1.2 Theoretical Basis of Multimodal Alignment

Alignment is a central concept in multimodal learning, referring to how models associate related information across modalities. Feature-level alignment deals with low-level characteristics like pixels or embeddings, while semantic alignment focuses on higher-level meaning. Many state-of-the-art models, such as CLIP and ALIGN, achieve alignment using contrastive learning, which trains models to bring semantically similar pairs closer in the embedding space while pushing dissimilar pairs apart. However, such approaches depend on large-scale paired or weakly paired datasets, which limits scalability and prevents generalization to unpaired real-world data where no explicit correspondence exists.

2.1.3 Architectural Foundations

In multimodal learning, alignment is a central concept that describes how models associate related information across modalities. Feature-level alignment deals with low-level properties such as pixels or embeddings, while semantic alignment concentrates on meaning at a higher level. By training models to push dissimilar pairs apart in the embedding space and bring semantically similar pairs closer together, contrastive learning is used by many SOTA(state-of-the-art) models, including CLIP and ALIGN, to establish alignment.

2.1.4 Learning from Unpaired Data

Unpaired multimodal learning addresses the challenge of training models when explicit relationships between modalities are absent. Unlike parallel datasets such as MS-COCO or Flickr30k, unpaired data lack aligned supervision, making it difficult for models to learn meaningful associations. To overcome this, researchers have proposed generative and alignment-free approaches, including cycle-consistent frameworks and reconstruction-based methods, which learn cross-modal correspondences indirectly. These approaches attempt to bridge the gap between modalities using latent alignment objectives rather than explicit pairwise matching, making them more suitable for naturally unstructured real-world data. However, such approaches rely on weakly paired or large-scale paired datasets, which restricts scalability and limits adaptation to unpaired real-world data where no explicit correspondence exists.

2.1.5 Efficiency and Generalization in Multimodal Models

Large models and paired datasets have a high computational cost, which prevents the scalability of multimodal learning. Improving computational efficiency involves adopting parameter sharing, lightweight encoders, and knowledge distillation methods that preserve representational power while using fewer resources. The ability of a model to function well across unknown domains and data distributions is known as generalization. Robust multimodal models must adapt to noise, missing data, and varying modalities. For representations to be more durable and transportable

across various settings, methods like constructive regularization, domain generalization, and data augmentation are essential.

2.2 Review of Existing Research

Large Multimodal Models (LLMs) have drastically transformed the way AI systems work with various types of data, such as image, text, and audio, especially benefiting educational applications and content creation. Adversarial attacks create a serious security risk for these models. It takes advantage of the weakness in how the models learn and connect data across different modalities, particularly while working with unpaired data. Because in the case of unpaired data, biases and vulnerabilities are harder to control. The research analyze these security threats and proposes a solution to make multimodal systems safer and more reliable, highlighting the need to establish better frameworks that can deal with various unpaired data while maintaining steady performance [1].

Moreover, recent work like Drive Anywhere leverages multimodal foundation models for end-to-end autonomous driving, illustrating the potential of large-scale vision-language models to generalize in open-set and out-of-distribution situations. They introduced patch-wise feature extraction to maintain spatial reasoning, and their method uses language-augmented latent space simulations for data augmentation and policy debugging. Although the approach proves effective in enhancing the robustness of autonomous driving, it still relies on paired multimodal attributes and particular task supervision, which limits its ability to fully utilize vast amounts of unpaired data. This underlines the need for more coherent and data-efficient frameworks that can fuse and align unpaired image-text representations, which the thesis aims to do [2].

In order to overcome the challenges of network traffic classification, this research combined explainable AI (XAI) methods with multimodal and multitask deep learning techniques. Variants of their DISTILLER framework, such as DISTILLER-EMBEDDINGS, EARLIER, and EVOLVED, improved dependability and decreased model complexity, as well as strengthened interpretability without compromising accuracy. Their approach still depends on carefully selected labelled datasets and task-specific supervision, although it shows the significance of multimodal fusion for robust classification. This limitation highlights the significance of scalable multimodal methods that can learn from unpaired and less organized data [3].

Furthermore, a recent research proposed a VAE-driven multimodal deep learning system that combines structural echocardiographic data and chest X-rays for the early diagnosis of cardiac conditions like SLVH and DLV. In comparison to VisualBERT, CLIP, and other baselines, their model achieves better diagnostic accuracy by combining EfficientNetB3 with attention mechanisms (SE-Block and CBAM), transformer encoders for structural input, and VAEs for latent space fusion. In the medical domain, it is highly effective; however, the technique based on paired multimodal datasets and curated health records, underlining the larger research gap of generalizable methods for learning from unpaired data [8].

Blockchain with multimodal data fusion improved data privacy and interpretability in health monitoring. The research approach uses Ethereum-based smart contracts, PBFT consensus, and decentralized storage (IPFS, Swarm) to ensure standard tamper-proof and standard fusion processes. However, the approach mainly relies on structural, paired medical datasets and mainly aims at security rather than generalization across different modalities. This research highlights the importance of secure, bias-free multimodal fusion and scalable alignment of unpaired multimodal data for broader application [9].

With the objective to enable a high-resolution and low-noise electrophysiological evaluation of 3D organoids, this recent research proposes a CMOS-based multimodal electrode array that can record, stimulate, and perform impedance spectroscopy across seven modalities. This research prioritizes sensor-level fusion over representation learning or cross-modal alignment. It also stresses hardware scalability and multimodal signal integrity. This reveals a gap in developing a learning framework that can efficiently describe unpaired and diverse multimodal data beyond hardware-level fusion and also highlights the importance of scalable multimodal unification [10].

The Multi-omics Trustworthy Integration Framework (MoTIF), the framework proposed by a recent research study, uses dynamic feature and modality selection along with estimation of uncertainty to improve interpretability and reliability in multimodal learning. The method combines early and late fusion techniques to dynamically allocate confidence to each modality, and it improves both prediction accuracy and reliability. However, scaling to unpaired or diverse modalities is not addressed by MoTIF; it depends mainly on paired multimodal omics data. Though this study points out a research gap in applying such dependable learning methods to unpaired multimodal datasets for wider generalization and underlines the importance of meaningful and dynamic alignment across modalities [11].

For multimodal knowledge graph completion, this research suggests a CLIP-enhanced Hyper-node Relational graph Attention Network (HRGAT), which incorporates textual, visual, and numerical information via low-rank multimodal fusion. The model effectively aligns image-text modalities and uses relation-specific attention to enhance forecasting missing links by using pre-trained CLIP embeddings. The approach shows increased ranking efficiency, particularly for head entities. This research leaves an opportunity to work on dynamic multimodal alignment and robust integration across various unstructured knowledge sources to increase scalability to unpaired data [12].

An interesting research proposes the EmoStream framework that presents a Multimodal Real-time Emotion Prediction System (MREPS) that combines physiological inputs, facial expression detection, and voice analysis into a reliable streaming architecture. A machine learning-driven emotion prediction API and speech-to-text converter improve dynamic classification accuracy in a variety of languages and scenarios. Although the system performs well for real-time multimodal fusion, it focuses mainly on synchronous data streams, which provides future opportunities in developing scalable, interpretable alignment for asynchronous multimodal data, boosting wider generalization and reliability in practical uses [13].

Another research suggests a parameter-efficient multimodal transfer learning model for Remote Sensing Visual Question Answering (RSVQA). The approach consists of a multimodal fusion module with self-attention and cross-attention mechanisms, a visual encoder initialized with weights pretrained on large-scale RSIs and a lightweight adaptor. Though, this approach successfully combines textual and visual information to achieve state-of-the-art performance on RSQV datasets, it still mostly relies on paired image-text data for training and uses different encoders for each modality. This research proposes a scope to develop a unified, scalable multimodal learning framework that can utilize unpaired data and mitigate augmentation-induced biases [14].

The study presents Multimodal Aligned Conceptual Knowledge (MACK), a knowledge-based framework for unpaired image matching that uses self-supervised bidirectional cycle-consistency to align words with prototypical vision region representation. The architecture complements cutting-edge models like CLIP and ALBEF by utilizing region-word similarity pooling and re-ranking techniques. Experimental results on MSCOCO and Flickr30k demonstrate notable gains in cross-dataset and zero-shot matching tasks. However, the work highlights various possibilities to improve multimodal alignment by pointing out weaknesses, including dependency on a non-SOTA detector (BUTD), limitation to nouns and adjectives, and absence of dynamic knowledge updates [15].

For the purpose of analyzing resilience against noisy or misaligned image-text pairs, this research links nonlinear contrastive loss to singular value decomposition (SVD), in order to provide a theoretical foundation for multimodal contrastive learning (MMCL). With the vision to improve representation recovery over unimodal self-supervised methods, this study suggest a semi-supervised extension of CLIP which enables the use of unpaired datasets via spectral graph partitioning and improved contrastive loss. The benefits of using unpaired data for improved feature alignment are represented by numerical tests on both synthetic and actual datasets. The work is limited to linear contexts and proof-of-concept trials, which highlights a research gap in scalable, non-linear multimodal architectures: developing reliable, unpaired multimodal representational learning techniques [16].

Attempting to capture cross-modal knowledge without spatial alignment, recent research presents a class-specific affinity-guided fully convolutional network (CSA-FCN) for unpaired multimodal image segmentation. It combines shared convolutional layers with modality-specific batch normalization and class-specific affinity matrices. This method notably outperformed UMMKD and single-modality baselines, achieving a standard Dice score of 87.65% (LGE) and 89.95% (bSSFP) on the MS-CMRSeg 2019 dataset. Similarly, it achieved a smaller Hausdorff distance across modalities on MM-WHS, reaching up to 96.07% Dice (MRI). Despite all of this progress, the framework only supports 2D feature maps and has no deeper interpretability for 3D extensions, leaving a research gap [17].

The existing multimodal AI systems mostly rely on paired datasets; therefore, the research offers a framework, MUM (Multi-Uni-Modal), that can handle both paired and unpaired data by using shared transformer encoders and task-specific heads with independent optimizers. Various cardiovascular inputs (ECG, echocardiogra-

phy, and phonocardiograms) are transformed into unified embeddings by the architecture, which leads to 16.9% improvement in the heart murmur detection task. Large-scale experiments show that MUM improves task performance by 2.7-4% while reducing 53.5% of learnable parameters. The research gap indicates a need to integrate continuous learning for dynamic clinical contexts with evolving pathologies and expand beyond cardiovascular diagnosis to other medical domains [18].

To overcome paired dataset dependency in multimodal medical AI, CLIP-IT pseudo-pairs unpaired TCGA pathology reports with histological images using CONCH-based CLIP retrieval. Knowledge distillation and LoRA- adapted encoders are used for text-to-vision translation with late fusion. The model outperforms all multimodal baselines in most cases, achieving 1.5-4% accuracy improvements on PCAM, BACH, and CRC datasets with little computational expense (0.006% FLOPs increase). The vulnerability towards external text quality and relevance (lower CRC performance with sparse reports), the lack of coordination with other modalities, and the limited exploration beyond classification tasks are research gaps that are specifically related to improving multimodal representation learning for unpaired data by enhancing semantic alignment [19].

The research addresses reinforcement learning with a Transformer architecture to handle unpaired multimodal neural machine translation. Sequence-level rewards are calculated in two ways such as the likelihoods of captioning models and attention weight cosine similarity between source-target language pairs grounded through shared images. In zero-resource translation circumstances, the method’s gains over baseline techniques on the Multi30k, IAPR-TC12, and IKEA datasets range from 1.3 to 2.0 BLEU points. Reliance on pre-trained captioning model quality, a lack of modern contrastive learning application, and a limited investigation of multimodal combinations other than visual-linguistic modalities are extending the opportunities to improve multimodal representation learning for unpaired data through better cross-modal alignment methods [20].

The core innovation of the paper "L-MCAT" is the Unpaired Multimodal Attention Alignment (U-mAA), which is specifically exciting as it directly addresses my objective of designing training strategies for truly unpaired data. It demonstrates how models can acquire cross-modal connections without explicit spatial correspondence or labels by integrating a contrastive self-supervised mechanism into the transformer attention. Additionally, the paper’s focus on lightweight architectural modifications, like Modality-Spectral Adapters, align perfectly with the thesis aim to improve computational viability and decrease hurdles, making sophisticated multimodal methodologies more accessible for research and deployment [21].

In order to address the data shortage, "Multimodal Data Augmentation for Image Captioning Using Diffusion Models" offers a crucial perspective. Their approach of using Stable Diffusion to produce excellent artificial image-caption pairs from text alone strikes me as especially perceptive. It shows that models trained on fully synthetic datasets can outperform traditional unpaired image captioning methods, achieving significant improvements, particularly in few-shot scenarios with limited labeled data. Our architectural enhancements focused at computational efficiency benefit greatly from the study’s multimodal augmentation approach, which com-

bines synthetic image generation with caption expansion through paraphrasing. Our strategy for sturdy training frameworks is driven by their results that suitable data filtering based on quality criteria (CLIPScore) can improve both training efficiency and performance [22].

Whenever ultrasound paired with microflow imaging samples are unavailable, GMRLNet, a graph-based manifold regularization learning framework addressing incomplete multimodal datasets in medical imaging. Its framework introduces a Graph Convolutional Network-based Shared and Specific Transfer Network (GSSTN) that separates modal inputs into interpretable shared and particular feature spaces. It concurrently captures global data distributions, local inter-sample relations, and sample-level features. Unlike GAN-based incomplete learning methods that generate pseudo images, GMRLNet achieves improved performance (0.913 AUC) even with 80 percent missing modality [23].

By merging structured tabular clinical records with unstructured discharge summaries, the paper Frattallone-Llado addresses temporal event extraction from healthcare data, highlighting the major benefits of multimodal approaches over text-only methods. Their annotation methodology supports our goal for enhanced multimodal representation learning by demonstrating that multimodal access reduces temporal uncertainty by 42%, 36%, and 13% for lower limits, upper bounds, and event durations, respectively. Importantly, their multimodal BERT architecture incorporates structured event data through multi-head attention mechanisms, performs 10% better than unimodal BERT and 80% better in F1 scores for bound predictions. Eventually, this validates our target of making architectural changes to foster cross-modal understanding. We emphasize on learning from truly unpaired scenarios, and the framework’s success with unpaired temporal relations where textual events lack explicit structured data timestamps is consistent with this goal [24].

By identifying and alleviating two types of data inconsistency: Modality-Modality (M-M) inconsistency, where information exists exclusively in one modality, and Modality-Label (M-L) inconsistency, where labels explain only one modality’s content, a crucial problem in multimodal learning. According to their validation, inconsistency occurrences are present in 63-76% of multimodal datasets, directly impacting retrieval performance. With a retrieval accuracy boost of up to 25.13%, the IA-CMR framework’s approach of disentangling modal representations into modality-common and modality-unique hash codes through contrastive learning is in line with our architectural enhancement objectives [25].

2.3 Summary of Key Findings

According to the literature study, multimodal learning has advanced significantly, but the majority of existing models rely largely on large-scale paired or weakly paired datasets, restricting their scalability and performance in real-world scenarios where such paired data is not available. In order to deal with unpaired data, some recent works have introduced methods like cycle-consistency, pseudo-pairing, and self-supervised learning to handle unpaired data. These approaches are often task-specific or computationally expensive.

Another recurring issue is modality inconsistency, which occurs when information is only available in one modality, or label inconsistency, which occurs when labels only describe a part of the data. Numerous studies also highlight how crucial computational efficiency is, as larger models struggle with high resource requirements. Lightweight architectures and knowledge distillation are gaining traction. Models that can form unstructured and unpaired data and manage real-world inconsistencies and generalize well are becoming more and more essential in a variety of fields, including knowledge of graphs, autonomous driving, and healthcare.

The main takeaway is that future research should concentrate on developing frameworks for learning from unpaired multimodal data that are more reliable, effective, and broadly applicable, including improvement in semantic alignment without relying on paired supervision, adopting lightweight models, and designing adaptable systems that can adapt to various data types and domains.

2.4 Work Management Plan

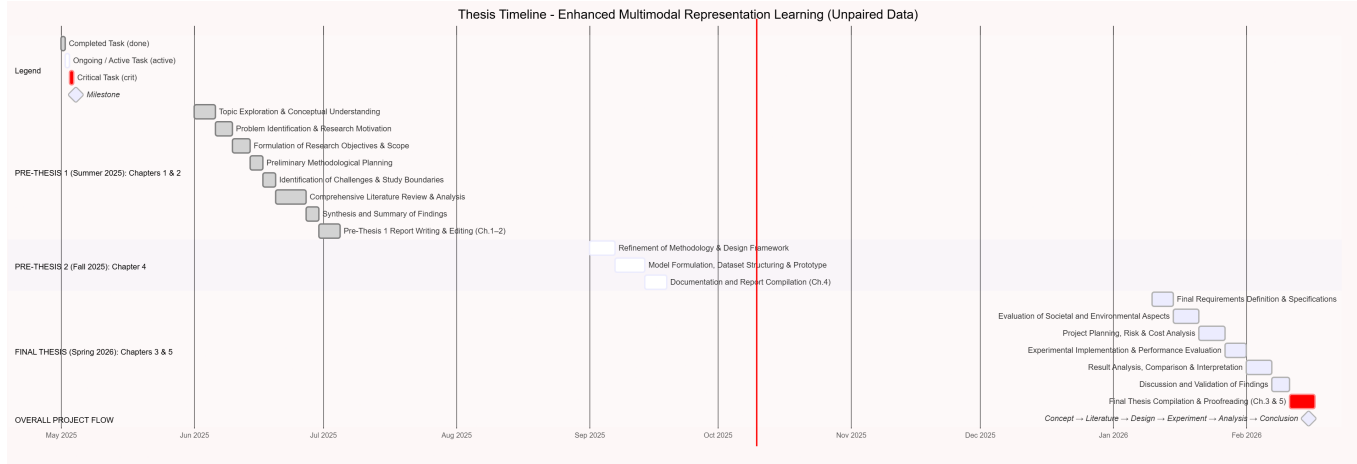


Figure 2.1: Gantt chart representing the structural development of the research paper on Enhanced Multimodal Representation Learning for Unpaired Data. From the introduction and literature review to methodology, requirements, results, and conclusion across Pre-Thesis 1, Pre-Thesis 2, and Final Thesis phases, it illustrates the thesis timetable.

2.5 Conclusion

A solid basis for the research into enhancing multimodal representation learning for unpaired data has been laid by the work done so far in Pre-Thesis 1, covering Chapter 1 and 2. It began with defining the motivation, problem statement, objective, and overall scope of the work, highlighting the difficulties faced by existing multimodal models that depends heavily on paired datasets. The introduction highlighted the significance of developing scalable, effective, and generalizable systems capable of learning from naturally unpaired sources. The literature review then provided a comprehensive understanding of existing multimodal learning approaches, such as models like CLIP and ALIGN, while identifying key limitations such as dependence

on costly parallel datasets, computational inefficiency, and lack of generalization. The roadmap for developing more reliable and accessible multimodal systems that can learn from unpaired data is outlined in this foundational work. As reflected in the Gantt chart, this phase, which wraps up the conceptual and theoretical stages, then gets ready for the following phase - model design and implementation in Pre-Thesis 2, and finally, the Final Thesis (Spring 2026) will focus on implementing the proposed framework, examining the outcomes, and assessing its practical usefulness and impact.

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